Chainage-Referenced Rock–TBM Spatial Interaction Graphs for Component-Level Diagnosis of Excavation Responses

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Abstract

TBM excavation is a moving rock–machine interaction process in which geological anomalies, machine geometry, and operating state jointly shape excavation responses. Monitoring curves can capture temporal response persistence, but they do not explicitly encode which geological units are spatially adjacent to which TBM components or how these relations evolve with chainage. This study proposes a chainage-referenced rock–TBM spatial interaction graph that formalises TSP-derived rock voxels and parameterized TBM surface samples as heterogeneous spatial entities. Candidate rock–machine relations are screened using active-zone membership, distance decay, surface-normal compatibility, and excavation-state constraints. Based on the resulting graph sequence, component–chainage diagnostic descriptors are derived by aggregating fixed-reference geological anomaly scores over geometry-weighted candidate relations. The descriptors are evaluated against persistence residuals of TBM monitoring responses using descriptor–residual association, descriptor-level null comparisons, detrending, circular-shift checks, and sensitivity analyses. Two tunnel cases indicate that the graph representation organises rock–machine relations into interpretable component–chainage patterns and provides diagnostic information not explicitly encoded by monitoring curves alone. The proposed descriptors are interpreted as geometry-weighted diagnostic summaries rather than measured contact forces, calibrated jamming probabilities, or universal operational risk rules.

Keywords: spatial representation; heterogeneous graph; rock–machine interaction; geometry-constrained relation graph; component–chainage descriptor; persistence residual; TBM excavation; diagnostic analysis
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1 Introduction

TBM excavation is not only a temporal monitoring process but also a moving spatial interaction process. As the machine advances, geological units ahead of the cutterhead and around the shield continuously enter and leave the potential rock–machine interaction zone. Abnormal thrust, torque, penetration, advance rate, or shield-pressure behaviour may therefore reflect not only temporal response persistence but also the evolving spatial configuration between geological anomalies and TBM components.

Existing TBM response studies commonly represent recent monitoring observations as a temporal sequence, $u_{t-K+1:t} \to r_{t+h}$, and can exploit the strong inertia of excavation responses [1, 11, 12, 28]. Advance geological prediction and TSP data provide a different kind of information: the spatial distribution of geological properties ahead of, or around, the excavation. These

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two data views remain weakly connected. A monitoring curve does not state which TSP voxel is adjacent to the cutterhead or shield, whether that adjacency is compatible with the local surface normal, or how the candidate rock–machine relation set changes as excavation proceeds.

This study treats the problem as a chainage-referenced spatial entity-relation modelling problem. TSP-derived rock voxels, parameterized TBM surface samples, and monitoring responses are placed in a unified local excavation coordinate system. At each excavation step, a heterogeneous graph $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$ represents rock voxels, machine components, rock–rock neighbourhoods, machine-surface adjacency, and geometry-constrained candidate rock–machine relations. The rock–machine edge set E_t^{rm} is not an arbitrary learned connectivity pattern; it is screened by active-zone membership, distance decay, surface-normal compatibility, and excavation state.

The central claim is therefore not that a new deep predictor outperforms persistence. Given the compact diagnostic intervals and the need for spatial interpretation, the more defensible question is whether a graph-based spatial relation representation encodes diagnostic information that monitoring curves and global geological summaries do not explicitly provide. The proposed component–chainage descriptors aggregate a fixed-reference TSP anomaly score over geometry-weighted candidate relations and are evaluated against persistence residuals, the part of the monitoring response not already explained by temporal inertia.

The contributions are threefold. First, the study proposes a chainage-referenced rock–TBM spatial entity-relation model that places TSP voxels, TBM surface components, and monitoring responses in a common excavation coordinate system. Second, it defines a geometry-constrained candidate rock–machine relation graph using active-zone filtering, distance decay, surface-normal compatibility, and excavation-state constraints. Third, it derives component–chainage diagnostic descriptors. The evaluation uses fixed descriptor–residual associations, descriptor-level null comparisons, detrending, and circular-shift checks. It also tests graph-threshold and anomaly-score sensitivity and retains edge-level traceability.

2 Related work

2.1 TBM response prediction and jamming-related analysis

Existing TBM prediction and warning studies have shown that tabular models and sequence models can be effective for forecasting penetration, thrust, torque, or jamming-related responses [1, 11, 28]. Sequence models are effective for learning temporal dependence in monitoring records [3, 12], but they usually treat excavation as an ordered signal sequence rather than as a spatially structured interaction process. As a consequence, existing studies usually represent geological and machine variables as feature vectors or sequences, rather than explicit spatial relations between geological entities and TBM components. This limitation matters when the objective is not only response prediction, but also spatially consistent interpretation of interaction dynamics.

2.2 Spatial representation of geological conditions in tunnel excavation

Tunnel seismic prediction (TSP) and advance geological prediction provide spatially distributed descriptions of the geological field ahead of or around excavation [14]. TSP-derived geological information can be transformed into chainage-referenced rock voxels that preserve spatial support and local geological context. Voxelised geological models and three-dimensional geological representation have been used to describe subsurface heterogeneity along tunnel alignments. Surface-based 3D geological models and broader 3D geospatial data models provide useful precedents for representing volumetric or surface-supported subsurface objects and linking them to other spatial entities [2, 10, 13, 17, 24]. This broader GIScience view is useful because tunnel geology, TBM geometry, and monitoring records have different spatial supports, and support mismatch can change the meaning of any aggregated descriptor [18]. However, this representation still needs to be linked to machine geometry: a geological voxel field alone does not specify which rock units are candidate interaction partners for which TBM surface components at a given excavation step. Bridging geological spatial representation and machine surface geometry is therefore a prerequisite for spatially explicit rock–machine interaction modelling.

2.3 Graph-based spatial relation modelling and spatial interaction descriptors

Graph concepts are useful for this problem because they provide a formal way to represent irregular spatial objects, contiguity, topology, and interaction. In GIScience, spatial relation models and graph-based data models have long been used to express neighbourhoods, topological relations, temporal dynamics, and geographical objects [4, 6, 8, 19, 24, 25]. Recent IJGIS and IJDE studies further show that graph-based representations can organise spatial units, neighbourhood structure, spatial multivariate distributions, and urban network morphology [5, 16, 32]. The important point for the present study is not that a graph must be used as a deep learning architecture, but that it can define an explicit spatial entity–relation data model.

Spatial interaction models also emphasise that relations are not arbitrary: they are shaped by distance decay, compatibility, direction, and spatial support [7, 23]. This logic is directly relevant to rock–TBM analysis, where a candidate rock–machine relation depends on active-zone membership, surface proximity, surface-normal compatibility, and excavation state. A graph therefore provides a disciplined representation of plausible candidate relations before any predictive model is considered.

Graph neural networks and heterogeneous graph networks remain relevant as possible extensions for learning on such structured data [22, 26, 29, 31], and graph-based forecasting models have been successful in other spatiotemporal domains [9, 15, 27, 30]. Graph networks have also been used to represent local interactions in physical systems and mesh-based simulation, which reinforces the value of explicit relation sets for spatially constrained processes [20, 21]. However, the present paper does not position the graph primarily as a black-box forecasting backbone. It uses the graph to derive auditable component–chainage spatial interaction descriptors from geometry-constrained rock–machine candidate relations.

2.4 Residual-based diagnostic evaluation

For short-horizon TBM monitoring, persistence is a strong and transparent temporal baseline. A spatial descriptor is therefore evaluated more appropriately by asking whether it co-varies with the response change that remains after persistence is removed, rather than whether it replaces temporal inertia. This residual view also reduces overclaiming: descriptor–residual association can support a spatial diagnostic interpretation without implying a calibrated forecasting model, a causal geological source, or an operational risk probability. In the present study, this logic is combined with descriptor-level null comparisons, detrending, circular-shift checks, and sensitivity analyses so that the reported evidence concerns the spatial relation representation rather than raw temporal trend alone.

3 Methodology

This section formalizes rock–TBM interaction as a chainage-referenced spatial entity-relation graph. The method has four steps: defining rock and machine spatial entities, constructing geometry-constrained candidate relations, aggregating component–chainage spatial interaction descriptors, and evaluating descriptor consistency with persistence residuals of monitoring responses.

3.1 Chainage-referenced rock–TBM spatial entity model

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \ldots, T$. The monitoring vector at step t is

$$u_t = [\mathrm{AdvanceRate}_t,\ \mathrm{RPM}_t,\ \mathrm{Torque}_t,\ \mathrm{Thrust}_t,\ \mathrm{Penetration}_t,\ \mathrm{ShieldPressure}_t],$$

and the response variables used for residual diagnostics are denoted $r_t^{(k)}$, where k indexes variables such as torque, thrust, penetration, advance rate, or shield pressure.

The geological input is a TSP-derived voxel field $D_{\mathrm{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ is the coordinate of rock voxel i and g_i is its geological attribute vector. The machine input is a parameterized TBM surface $M_{\mathrm{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial surface node coordinate, n_j is the local surface normal, and ρ_j is the machine component label. The current implementation partitions the machine surface into cutterhead, front shield, middle shield, and tail shield components.

Because TSP coordinates and TBM monitoring mileage are commonly recorded in different local frames, all entities are mapped to a common local chainage coordinate. A simple alignment is

$$x_{\mathrm{local}}(t) = \mathrm{ShieldMileage}(t) - \mathrm{ShieldMileage}_{\mathrm{start}},$$

after which TBM surface nodes are advanced along the excavation direction d:

$$p_j(t+1) = p_j(t) + \Delta x\, d.$$

Table 1 summarises the spatial data model. The table is included to make clear that the graph is a queryable entity-relation representation, not only a plotting device or a neural-network input tensor.

Table 1: Spatial entities and relations in the chainage-referenced rock–TBM graph.

Object	Definition	Diagnostic role
V_t^r	Active TSP-derived rock voxels	Geological spatial support
V_t^m	Parameterized TBM surface samples	Cutterhead and shield component support
E_t^{rr}	Rock–rock voxel neighbourhoods	Geological contiguity context
E_t^{mm}	TBM surface neighbourhoods	Machine-surface topology context
E_t^{rm}	Screened rock–machine candidate relations	Geometry-compatible interaction support
$A_c(t)$	Sum of geometry weights for component c	Supporting exposure descriptor
$I_c(t)$	Geometry-weighted anomaly intensity for component c	Main diagnostic descriptor
$e_{t+h}^{(k)}$	Persistence residual of response k	Response-consistency target

3.2 Geometry-constrained rock–machine candidate relation graph

At each excavation step, a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m,\ E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

where V_t^r and V_t^m are rock and TBM surface node sets, E_t^{rr} denotes rock–rock voxel adjacency, E_t^{mm} denotes TBM surface adjacency, and E_t^{rm} denotes candidate rock–machine relations.

Rock node features are

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether voxel i remains active and eligible for interaction at step t. TBM node features are

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

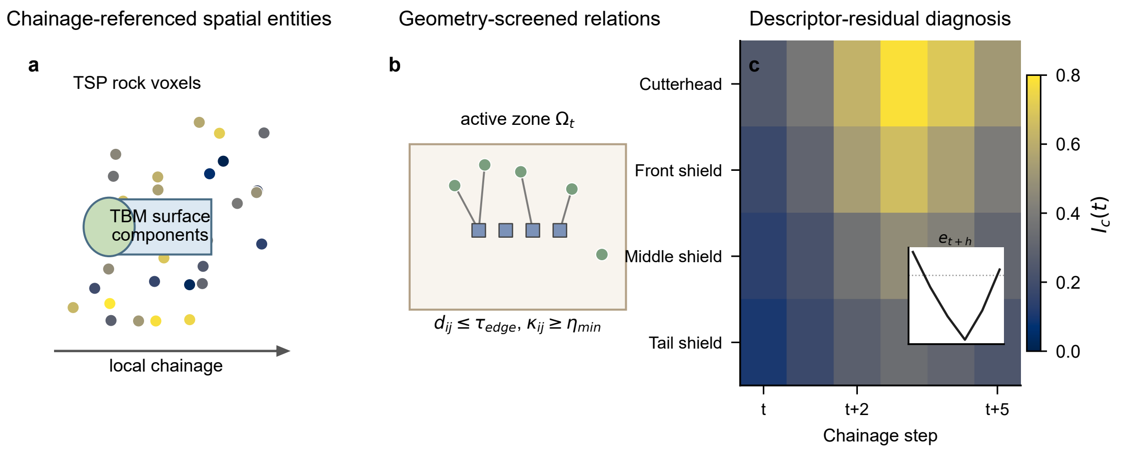

Figure 1: Overview of the proposed chainage-referenced rock–TBM spatial interaction graph workflow. The main manuscript focuses on spatial entities, geometry-constrained candidate relations, component–chainage descriptors, and response-residual consistency. The descriptors are not contact pressure, contact force, jamming probability, or calibrated risk.

The intra-rock edge set E_t^{rr} is constructed from voxel neighbourhoods, and E_t^{mm} is constructed from local neighbourhoods on the parameterized TBM surface. The key graph-design decision is the construction of E_t^{rm}.

A candidate edge (i, j) is included in E_t^{rm} only when four conditions are satisfied:

$$\begin{cases} d_{ij}(t) \leq \tau_{\text{edge}}, \\ \kappa_{ij}(t) \geq \eta_{\text{min}}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

where

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the distance between rock voxel i and TBM surface node j, and

$$\kappa_{ij}(t) = \max\left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon}\right)$$

is the signed-normal compatibility. The active zone Ω_t restricts candidate relations to the excavation-relevant rock volume, while $s_i(t)$ prevents already excavated voxels from repeatedly contributing to later graph snapshots. It is defined as the union of the face look-ahead zone and the shield-adjacent zone:

$$\Omega_t = \Omega_t^{\text{face}} \cup \Omega_t^{\text{shield}}.$$

With $x_f(t)$ denoting the cutterhead chainage, $x_{\text{tail}}(t)$ the tail shield chainage, $r_i = \sqrt{y_i^2 + z_i^2}$, cutterhead radius R_c, shield radius R_s, face look-ahead length L_f, and radial active-zone buffer τ_{zone}, the two parts are

$$\Omega_t^{\text{face}} = \{i \mid 0 \leq x_i - x_f(t) \leq L_f,\ r_i \leq R_c + \tau_{\text{zone}}\},$$

$$\Omega_t^{\text{shield}} = \{i \mid x_{\text{tail}}(t) \leq x_i \leq x_f(t),\ R_s \leq r_i \leq R_s + \tau_{\text{zone}}\}.$$

This definition separates face-contact plausibility from shield-side proximity while keeping both parts within one active rock volume for graph construction.

Table 2: Active-zone and candidate-relation parameters used in all formal case runs.

Parameter	Value	Role
L_f	5.0 m	Face look-ahead length for Ω_t^{face}
τ_{zone}	5.0 m	Radial active-zone buffer around face and shield
R_c	4.0 m	Parameterized cutterhead radius
R_s	3.95 m	Parameterized shield radius
Shield length	10.0 m	Axial support for Ω_t^{shield}
τ_{edge}	2.0 m	Candidate-edge distance threshold in the main setting
$\eta_{\min}$	0.3	Minimum surface-normal compatibility in the main setting
Surface sampling	0.5 m	TBM surface-node sampling resolution

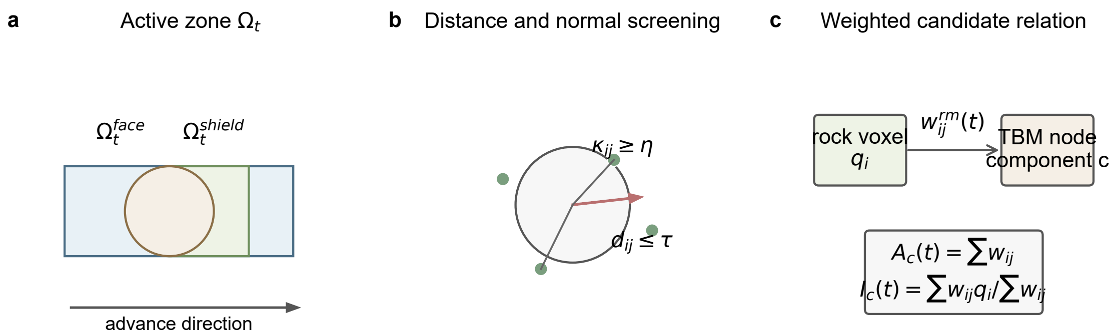

Figure 2: Geometry-constrained rock–machine candidate relation construction. Candidate relations are controlled by active-zone membership, distance decay, surface-normal compatibility, and excavation state. The construction defines spatially plausible relations; it does not recover measured contact.

For each candidate relation, the geometric interaction weight is defined as

$$w_{ij}^{rm}(t) = \exp\left(-\frac{d_{ij}(t)}{\tau_{\text{edge}}}\right)\kappa_{ij}(t).$$

This weight is the explicit form of geometry-compatible interaction intensity used by the main descriptors. It increases when a rock voxel is closer to the TBM surface and more compatible with the local surface normal. It remains a geometric descriptor, not a contact force or pressure.

3.3 Component–chainage spatial interaction descriptors

The geological contribution of each rock voxel is represented by a restrained TSP-derived anomaly score

$$q_i = \phi(g_i).$$

In the main formulation, q_i is a bounded low-velocity anomaly score derived from the training active-zone distribution of P-wave velocity:

$$q_i = \text{clip}\left(\frac{Q_{95}^{\text{train}}(Vp) - Vp_i}{Q_{95}^{\text{train}}(Vp) - Q_5^{\text{train}}(Vp) + \epsilon}, 0, 1\right).$$

This fixed training-reference scaling avoids per-chainage min–max normalisation and keeps descriptor values comparable along chainage. Lower V_p values therefore correspond to higher geological anomaly scores under a common reference distribution. This single-attribute definition is intentionally transparent and reproducible across the BSLL and SJLS cases. Multi-attribute variants can be examined as sensitivity checks, but they are not required for the main descriptor.

For each TBM component c, let V_c^m denote the set of TBM surface nodes whose component label is c. Two component–chainage descriptors are then defined. The pure geometric exposure is

$$A_c(t) = \sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t),$$

where $A_c(t)$ is a supporting total geometric exposure descriptor. It is affected by component surface extent, surface-node sampling density, candidate-edge count, and active-zone geometry. It is therefore used mainly to document total candidate-relation exposure; when direct cross-component comparison is required, a node-normalised exposure

$$\bar{A}_c(t) = \frac{A_c(t)}{|V_c^m| + \epsilon}$$

can be used as a sensitivity form. The geology-weighted spatial interaction intensity is

$$I_c(t) = \frac{\sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t) q_i}{\sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t) + \epsilon}.$$

$A_c(t)$ describes how strongly component c is geometrically exposed to the screened candidate rock–machine relation set, whereas $I_c(t)$ describes the weighted geological anomaly intensity around that component. These descriptors are defined for cutterhead, front shield, middle shield, and tail shield at each chainage step, producing a component-by-chainage diagnostic map.

3.4 Response-residual consistency analysis

Persistence is a strong baseline for short-horizon TBM monitoring responses. The proposed graph descriptors therefore do not attempt to replace temporal inertia. Instead, they are evaluated against persistence residuals:

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)},$$

where h is the diagnostic horizon and k indexes a response variable. The diagnostic question is whether component–chainage descriptors co-vary with the remaining response change after persistence has been removed:

$$I_c(t) \to e_{t+h}^{(k)}.$$

The formal reported statistic is Spearman correlation between $I_c(t)$ and $e_{t+h}^{(k)}$ over the evaluated test chainage interval. Spearman correlation is used because the diagnostic question concerns monotonic descriptor–residual co-variation under small sample sizes, rather than a calibrated linear response model. The result is interpreted as descriptor–response residual co-variation within the evaluated chainage interval, not as causal attribution.

3.5 Null-model, sensitivity, and traceability checks

The diagnostic descriptor–residual pairs are specified before the null-model and sensitivity checks: BSLL h=1 uses front-shield $I_c(t)$ against advance-rate residuals, BSLL h=3 uses front-shield $I_c(t)$ against shield-pressure residuals, and SJLS h=3 uses cutterhead $I_c(t)$ against shield-pressure residuals. These pairs serve as representative diagnostic slices covering a compact one-step BSLL interval, a multi-step BSLL shield-pressure interval, and the external SJLS tunnel case. They are not re-selected during null comparison or sensitivity analysis; the full exploratory component–response matrix is reported separately in Appendix A.

Five descriptor-level null variants are used to test whether the proposed descriptor is reducible to simpler summaries. The chainage-only variant uses target chainage as the descriptor.

The global-Vp variant uses the mean low-velocity anomaly score over active rock voxels, without component-specific geometry. The distance-only variant uses $A_c(t)$ and therefore contains geometry exposure but no geological anomaly weighting. The uniform-edge variant replaces geometry-weighted averaging with an unweighted mean anomaly over candidate edges. The component-reassignment null uses the same graph construction but randomly assigns the comparison to non-target TBM components with a fixed random seed, breaking the intended component label without claiming a formal permutation test.

Two small-sample robustness checks are then applied to each fixed diagnostic pair. First, both descriptor and residual series are linearly detrended against chainage, and Spearman correlation is recomputed on the detrended series. Second, a circular-shift check rolls the residual sequence by all non-zero shifts and reports the proportion of shifted correlations whose absolute value is at least as large as the observed value. These values are descriptive diagnostics rather than confirmatory p-values.

To account for TSP–TBM chainage alignment uncertainty, the fixed diagnostic descriptor series is also shifted by $\delta_x \in \{-1, 0, +1\}$ m using linear interpolation along target chainage, and raw and detrended Spearman correlations are recomputed. This test does not replace a full geophysical registration analysis, but it checks whether the reported diagnostic pattern depends on exact metre-level alignment.

Anomaly-definition sensitivity is evaluated by replacing the main Vp-based score with Vs-low, Vp/Vs-low, and mean(Vp, Vs) low-anomaly variants. For traceability, candidate edges for a fixed diagnostic sample are ranked by their contribution $w_{ij}^{rm}(t)q_i$, retaining the rock voxel, TBM node, distance, normal compatibility, geometry weight, anomaly score, and weighted contribution.

4 Case studies and experimental design

4.1 Case roles and data sources

The experiments use two real tunnel cases with complementary evidential roles. The BSLL tunnel case starts from DyK1017+205 and is evaluated under two diagnostic horizons: h=1 and h=3. It provides a compact case for testing whether component–chainage descriptors can be constructed and related to response residuals over short target intervals. The SJLS tunnel case starts from Dyk1252+411 and is evaluated under h=3, providing the clearest external TSP contrast with a longer test interval for descriptor–residual association.

For each excavation step, the monitoring vector contains advance rate, cutterhead RPM, torque, thrust, penetration, and shield pressure. The diagnostic responses are advance rate, torque, thrust, penetration, and shield pressure. The geological input is a TSP-derived voxel field. For BSLL, the project TSP voxel field contains P-wave velocity, S-wave velocity, density, dynamic Young's modulus, velocity ratio, and Poisson's ratio around the tunnel alignment. For SJLS, external P-wave and S-wave velocity profiles at Dyk1252+411 are preprocessed into the same project TSP schema and paired with mileage-aligned monitoring records.

4.2 Samples and graph-construction settings

All cases use mileage-ordered partitioning. Descriptor values are computed from the last observed graph snapshot of each target sample, and the last observed response at step t is used to define the persistence residual for a target response at $t + h$. The formal case runs are summarised in Table 3.

The BSLL one-step run has target chainages 41–48 m in the test interval. The BSLL three-step run has target chainages 42–48 m. The SJLS run has target chainages 99–115 m. The graph construction uses $\tau_{\text{edge}} = 2.0$ m, $\eta_{\min} = 0.3$, and $\tau_{\text{zone}} = 5.0$ m in the main setting. The

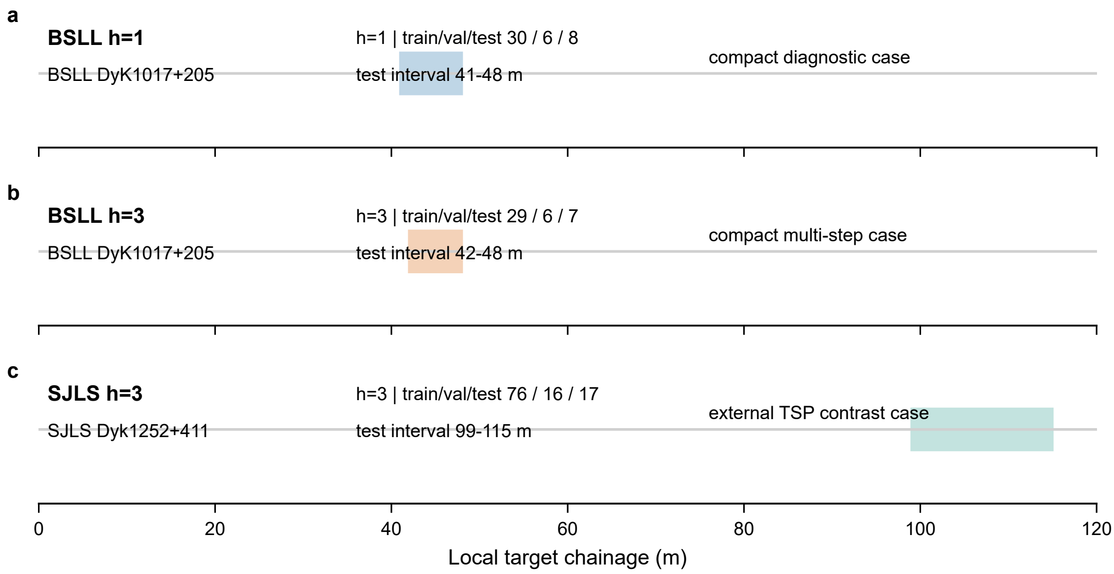

Figure 3: Real tunnel scenarios used to instantiate the common descriptor framework. BSLL DyK1017+205 provides compact h=1 and h=3 cases, while SJLS Dyk1252+411 provides the clearest external TSP descriptor evidence under h=3.

Table 3: Formal case runs used in the descriptor experiments. Effective samples are chainage-indexed residual samples after applying the diagnostic horizon.

Case	Tunnel	Start chainage	Horizon	Train/Val./Test samples
BSLL h=1	BSLL	DyK1017+205	1	30 / 6 / 8
BSLL h=3	BSLL	DyK1017+205	3	29 / 6 / 7
SJLS h=3	SJLS	Dyk1252+411	3	76 / 16 / 17

TBM surface is discretised into cutterhead, front-shield, middle-shield, and tail-shield surface nodes at a resolution of 0.5 m.

Table 4: TBM geometry and surface sampling settings used in graph construction.

Case	Cutterhead radius (m)	Shield radius (m)	Front (m)	Middle (m)	Tail (m)	TBM nodes
BSLL h=1	4.0	3.95	3.0	3.5	3.5	1352
BSLL h=3	4.0	3.95	3.0	3.5	3.5	1352
SJLS h=3	4.0	3.95	3.0	3.5	3.5	1352

All formal runs use a 0.5 m surface sampling resolution, so the strongest SJLS diagnostic evidence is not produced under a coarser TBM surface discretisation than the BSLL cases.

TSP preprocessing follows a common voxel schema. Coordinates are converted to a local excavation coordinate by subtracting the minimum longitudinal coordinate, so the local X axis is aligned with advance direction. Each voxel can store $[V_p, V_s, E, V_p/V_s, \nu, \rho]$ when these attributes are available, and available attributes are standardised using statistics fitted on the training active-zone margin. This multi-attribute schema keeps the BSLL and SJLS inputs in a common voxel format. However, the main spatial interaction descriptor uses only the cross-case consistent P-wave velocity attribute. The geological anomaly score is therefore a $[0, 1]$ low-velocity anomaly score derived from the training active-zone reference distribution of V_p; derived or case-specific attributes such as V_s, modulus, velocity ratio, Poisson's ratio, and density are not used in the main descriptor calculation.

4.3 Diagnostic evaluation metrics

The diagnostic outputs are component–chainage descriptors. For each component c, $A_c(t)$ is a supporting exposure descriptor that reports the sum of geometry weights over candidate relations, and $I_c(t)$ is the main geology-weighted descriptor. The response consistency target is the persistence residual $e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}$ for each response variable k.

Descriptor–residual association is formally reported with Spearman correlation between $I_c(t)$ and the persistence residual. A threshold sensitivity analysis varies τ_{edge} and $\eta_{\min}$ to check whether the descriptor–residual pattern is tied to one specific graph-construction setting. Null-model comparisons, detrending, circular-shift checks, anomaly-definition sensitivity, and edge-level traceability are applied to the fixed diagnostic pairs. The results are interpreted as chainage-local diagnostic co-variation, not as causal evidence or calibrated operational risk.

5 Results

5.1 Constructed rock–machine interaction graph sequences

The first result is the constructed spatial relation object itself. For each case, TSP-derived rock voxels and parameterized TBM surface nodes are mapped to the same chainage-referenced coordinate system, and the candidate relation set E_t^{rm} is rebuilt at each excavation step. Figure 2 illustrates the construction rule: candidate relations are screened by active-zone membership, distance threshold, surface-normal compatibility, and excavation state.

Table 5 summarises the resulting graph objects in the test intervals. The BSLL runs use the same tunnel section with different diagnostic horizons and therefore have the same graph size per step. All three formal runs use the same 0.5 m TBM surface sampling resolution, so the SJLS evidence is not produced under a coarser surface discretisation. The candidate-edge counts show that the descriptors are computed from explicit rock–machine relation sets rather than from conceptual links. Because the same parameterized TBM surface, active-zone dimensions, and edge thresholds are used in the three formal runs, the graph-size statistics are identical.

The case contrast is therefore expressed by the geological anomaly scores and by the resulting component descriptors, not by a different graph size.

Table 5: Graph construction statistics in the test intervals. Edge shares are computed from candidate rock–machine relations in the main graph setting.

Case	Rock nodes	TBM nodes	Candidate edges	Cutterhead share	Shield share
BSLL h=1	3558	1352	14620	0.145	0.855
BSLL h=3	3558	1352	14620	0.145	0.855
SJLS h=3	3558	1352	14620	0.145	0.855

Table 6: Case-level descriptor attribute statistics. Mean anomaly is the candidate-edge mean of the bounded low-velocity score q_i; $I_c(t)$ is the geometry-weighted anomaly descriptor pooled over components and test chainages.

Case	Mean anomaly score	$I_c(t)$	$A_c(t)$
BSLL h=1	0.552 ± 0.157	0.552 ± 0.158	1170.8 ± 296.9
BSLL h=3	0.537 ± 0.129	0.537 ± 0.130	1170.9 ± 297.4
SJLS h=3	0.913 ± 0.045	0.913 ± 0.045	1170.9 ± 294.2

5.2 Component–chainage descriptor patterns and residual traces

The main graph-derived output is the component–chainage descriptor map built from $I_c(t)$. In each case, cutterhead, front-shield, middle-shield, and tail-shield values are computed across the test chainage interval as geometry-weighted TSP anomaly summaries over candidate rock–machine relations.

Figure 4 links these descriptor maps to the fixed diagnostic residual traces. The heatmaps show where low-velocity anomaly exposure is concentrated by component and chainage, while the lower panels show the corresponding response residual selected for diagnostic checking. This layout makes the evidence chain explicit: the graph first produces a spatial component–chainage descriptor, and the residual trace then tests whether that descriptor co-varies with response changes left after persistence.

5.3 Representative diagnostic descriptor–residual associations

The formal diagnostic statistic is Spearman correlation between $I_c(t)$ and the persistence residual $e^{(k)}_{t+h}$. Table 7 reports the representative diagnostic pairs together with raw Spearman correlation, linearly detrended Spearman correlation, and the circular-shift diagnostic. One pair is fixed for each case to prevent parameter-wise re-selection during null and sensitivity checks. The pairs cover the compact BSLL h=1 interval, the BSLL h=3 shield-pressure interval, and the external SJLS shield-pressure case. The values are descriptive and are not treated as confirmatory hypothesis tests.

Table 7: Representative diagnostic descriptor–residual diagnostics for fixed component–response pairs. Circular-shift values are descriptive small-sample checks.

Case	Diagnostic pair	n	Raw ρ	Detrended ρ	Shift check
BSLL h=1	Front shield–advance rate	8	0.381	0.214	0.375
BSLL h=3	Front shield–shield pressure	7	-0.750	-0.464	0.429
SJLS h=3	Cutterhead–shield pressure	17	-0.912	-0.772	0.059

The SJLS case shows the most evident descriptor–residual co-variation. Its cutterhead $I_c(t)$–shield-pressure residual association remains strong after linear detrending ($\rho = -0.772$),

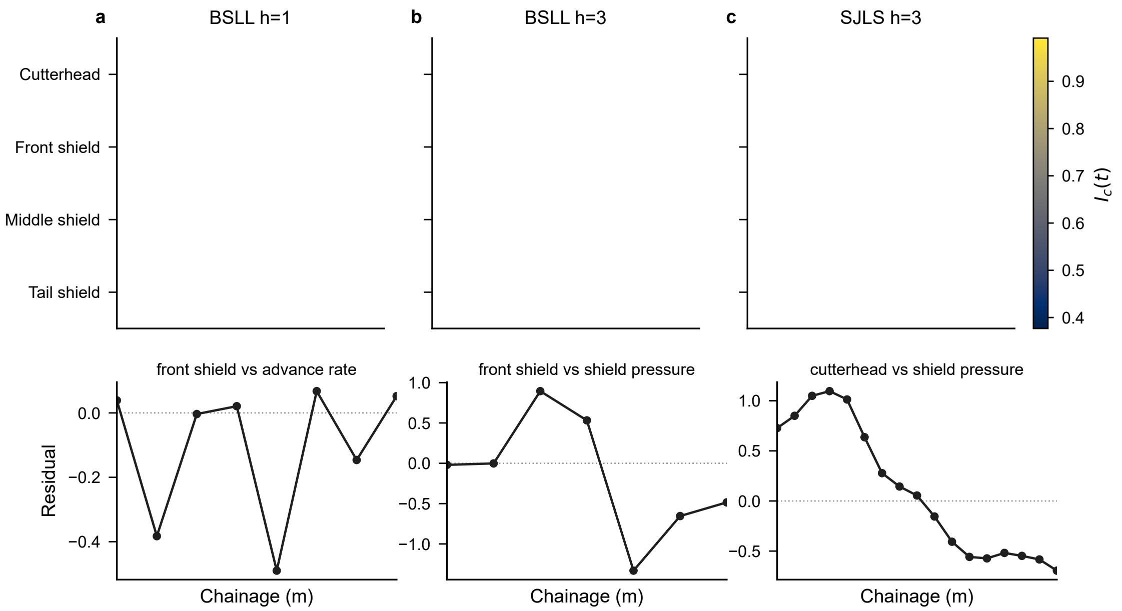

Figure 4: Component–chainage spatial interaction descriptor evidence and diagnostic residual traces. Upper panels show $I_c(t)$ values for BSLL h=1, BSLL h=3, and SJLS h=3 using a common colour scale. Lower panels show the fixed residual series used for the fixed diagnostic pair in each case.

whereas the BSLL evidence weakens under detrending. This difference is consistent with the longer SJLS test interval and clearer external TSP velocity contrast. The BSLL results are retained as compact diagnostic checks, but they do not support a strong case-general rule.

For completeness, Appendix A reports the full component–response Spearman matrix for $I_c(t)$. The matrix is exploratory and unadjusted for multiple testing. The representative diagnostic pairs in Table 7 are the main result used in the text.

5.4 Descriptor-level null comparisons

The null-model comparison tests whether the fixed diagnostic associations can be explained by simpler summaries. Figure 5 shows raw and detrended correlations for the proposed descriptor and five null variants. The most important diagnostic is the detrended comparison, because chainage and global anomaly can both carry strong monotonic trends in short test intervals.

Table 8: Detrended Spearman correlations for the proposed descriptor and key null variants.

Case	Proposed	Chainage	Global Vp	Distance only	Component reassign.
BSLL h=1	0.214	0.000	-0.190	0.000	0.071
BSLL h=3	-0.464	-0.145	-0.321	0.000	0.286
SJLS h=3	-0.772	-0.009	-0.179	0.000	0.130

For SJLS, the raw chainage-only and global-Vp associations are high, but both collapse after detrending. The proposed descriptor retains a strong detrended association, indicating that the component-specific geometry-weighted relation representation contains more than a simple chainage trend. The random component-reassignment null is also much weaker than the proposed SJLS descriptor after detrending. The BSLL null comparisons are weaker and are read as limited-sample diagnostics.

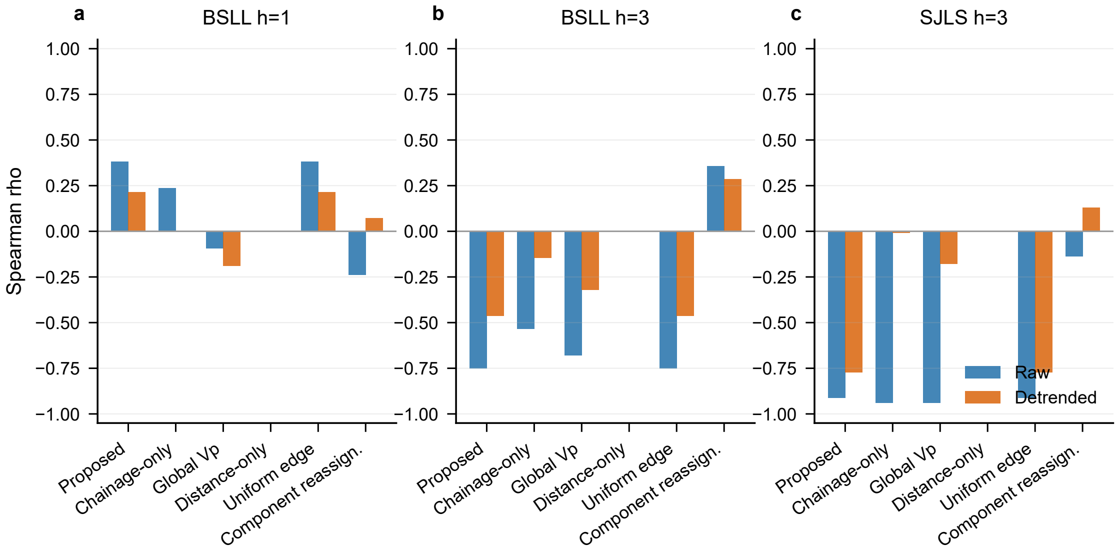

Figure 5: Descriptor-level null-model comparison for the representative diagnostic pairs. Bars report raw and linearly detrended Spearman correlations for the proposed descriptor and five simpler variants. The comparison evaluates descriptor diagnostic structure, not prediction accuracy.

5.5 Sensitivity to graph thresholds and anomaly definitions

Figure 6 tests whether the fixed $I_c(t)$–residual pairs are preserved when the edge-distance threshold τ_{edge} and normal-compatibility threshold $\eta_{\min}$ are varied. The same descriptor–response pair is held fixed in each case, so the figure evaluates threshold stability rather than searching for a new strongest pair.

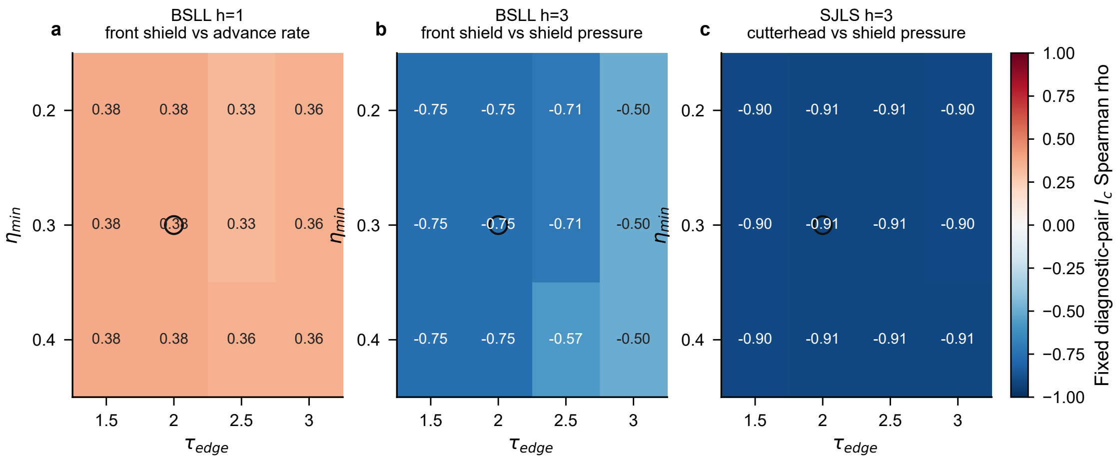

Figure 6: Sensitivity of fixed $I_c(t)$–residual Spearman pairs to graph-construction thresholds. Each panel varies τ_{edge} and $\eta_{\min}$ for one formal case while holding the descriptor–response pair fixed.

Table 9 reports the anomaly-definition sensitivity for the same fixed diagnostic pairs. SJLS remains strong under Vp, Vs, and mean(Vp, Vs) low-anomaly definitions after detrending. BSLL is more sensitive to the anomaly definition, which is consistent with its compact test intervals.

Table 10 reports the TSP–TBM chainage alignment uncertainty check. The SJLS association remains negative under $\pm 1\,\text{m}$ offsets and remains strong after detrending at the 0 and $+1\,\text{m}$

Table 9: Detrended Spearman correlations under alternative anomaly-score definitions for the fixed diagnostic pairs.

Case	Vp main	Vs low	Vp/Vs low	Mean(Vp, Vs)
BSLL h=1	0.214	-0.214	0.214	-0.024
BSLL h=3	-0.464	0.179	-0.179	-0.643
SJLS h=3	-0.772	-0.873	0.338	-0.863

settings. The BSLL checks are more variable, again reflecting their short test intervals.

Table 10: Sensitivity of fixed diagnostic pairs to ±1 m longitudinal descriptor alignment offsets. Values report raw/detrended Spearman correlations.

Case	−1 m	0 m	+1 m
BSLL h=1	0.250 / -0.107	0.381 / 0.214	-0.071 / 0.250
BSLL h=3	-0.143 / 0.257	-0.750 / -0.464	-0.657 / -0.714
SJLS h=3	-0.850 / -0.397	-0.912 / -0.772	-0.938 / -0.874

5.6 Edge-level traceability

The graph representation also supports edge-level inspection. For the SJLS diagnostic pair, Figure 7 shows the top contributing candidate rock–machine relations at the selected chainage. Each contribution is traceable to a rock voxel, a TBM surface node, a distance, a normal compatibility value, a geometry weight, and a low-velocity anomaly score.

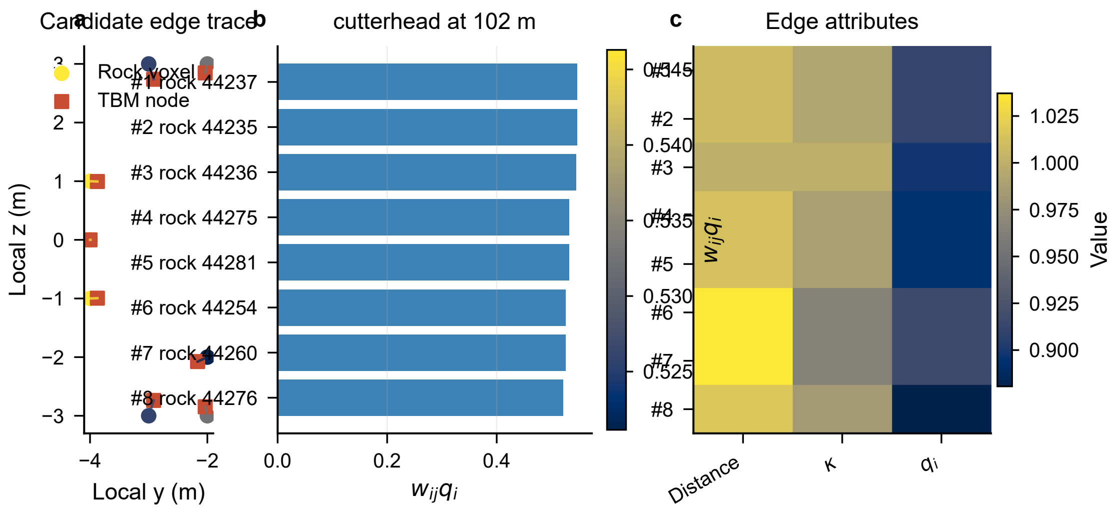

Figure 7: Traceability example for a fixed diagnostic pair. Candidate rock–machine edges are ranked by $w_{ij}^{rm}(t)q_i$, showing how the component–chainage descriptor can be traced back to individual rock voxels and TBM surface nodes.

This traceability is the main practical difference between the proposed graph descriptor and a monitoring-only residual curve. The descriptor can be queried back to the spatial relation set that produced it, while the residual curve alone cannot identify which geological voxels and machine components were involved in the diagnostic summary.

6 Discussion

6.1 Spatial data-model contribution

The main contribution is not a higher-accuracy response predictor, but a chainage-referenced spatial relation model that makes the otherwise implicit rock–machine relation set explicit, inspectable, and queryable. Monitoring sequences can describe response inertia, and TSP voxels can describe geological heterogeneity, but neither representation alone states which geological voxels are candidate interaction partners for which TBM components at a given chainage. The proposed graph closes this representational gap by linking rock voxels, TBM surface nodes, geometry-compatible candidate edges, and monitoring residuals in one excavation coordinate system.

This distinction matters for IJGIS/IJDE-style spatial modelling. The graph is not introduced merely as a deep-learning architecture. It is a spatial data model with explicit entities, relation types, screening rules, edge weights, component aggregation, and traceability back to individual candidate relations. The descriptor results are therefore interpreted as evidence that the representation can organise rock–machine relations into component–chainage diagnostic summaries.

6.2 Diagnostic meaning beyond monitoring curves

Comparing $I_c(t)$ with persistence residuals reframes the evaluation question from "does the graph beat persistence?" to "does the graph encode spatial diagnostic information consistent with the response changes left after persistence?" This is a more defensible question for the present data scale. The SJLS results illustrate the value of this framing: raw chainage and global Vp summaries are strongly associated with the selected residual, but their associations largely disappear after linear detrending, whereas the proposed component-specific descriptor remains strong.

The BSLL results are more limited. They show that the graph and descriptors can be constructed consistently in compact intervals, but the residual-consistency evidence weakens after detrending and changes with anomaly definition. This does not invalidate the graph construction; rather, it defines the appropriate claim boundary. The evidence supports a diagnostic spatial-representation claim, not a universal component–response law.

6.3 Case-specific evidential strength

The contrast between SJLS and BSLL is not treated as a success–failure dichotomy. It shows that descriptor usefulness depends on geological contrast, monitoring quality, operational stability, and the length of the evaluated chainage interval. SJLS contains a longer test interval and a clearer external TSP velocity contrast, which makes descriptor–residual co-variation more visible. BSLL h=1 and h=3 contain only 8 and 7 test samples, respectively, and their residuals may be more strongly affected by operational control and local construction variability.

The circular-shift checks are also read conservatively. They are useful as small-sample diagnostics for monotonic sequence structure, but they are not formal multiplicity-adjusted significance tests. The strongest evidence is the combined pattern: fixed diagnostic pairs, detrended comparison, descriptor-level null models, threshold sensitivity, anomaly-definition sensitivity, and edge-level traceability.

6.4 Interpretation boundary

The proposed descriptors are geometry-weighted summaries of TSP-derived anomalies around TBM components. They are not measured contact force, contact pressure, jamming probability, causal attribution, or calibrated operational risk. This boundary is important because the graph encodes candidate spatial relations, not observed physical contact. Independent engineering labels would be required before the descriptors could be interpreted as risk probabilities or contact-force estimates.

6.5 Limitations and future work

Several limitations remain. First, the current evidence comes from two tunnel cases and short diagnostic test intervals, so additional tunnel sections and geological conditions are required for generalisation. Second, independent engineering evidence such as geological exposure logs, jamming records, shield pressure anomaly logs, and construction intervention records is needed to validate operational interpretations. Third, TSP inversion uncertainty is not propagated through the voxel graph; future work needs uncertainty-aware voxel attributes and evaluate how spatial resolution affects candidate relation sets.

Fourth, the main descriptor intentionally uses a transparent Vp-based low-velocity anomaly score. Multi-attribute scores using Vs, modulus, Vp/Vs, Poisson's ratio, or density require clear weighting rules or independent validation. Finally, with larger samples, the full heterogeneous graph structure E^{rr}, E^{mm}, and E^{rm} could be used for learned representation models. Such extensions remain secondary to the spatial entity-relation claim unless sufficient training and validation data become available.

7 Conclusion

This study develops a chainage-referenced spatial entity-relation framework for rock–TBM interaction diagnosis. The framework places TSP-derived rock voxels, parameterized TBM surface components, and monitoring responses in a common excavation coordinate system, so that geological units, machine components, and response residuals can be analysed through explicit spatial relations.

The framework constructs geometry-constrained candidate rock–machine relations using active-zone membership, distance decay, surface-normal compatibility, and excavation-state constraints. It then derives component–chainage diagnostic descriptors from geometry-weighted TSP anomaly summaries. Descriptor-level null comparisons, detrending, circular-shift checks, threshold sensitivity, anomaly-definition sensitivity, and edge-level traceability are used to evaluate whether these descriptors contain residual-consistent diagnostic information.

The two case studies indicate that the descriptors can organise rock–machine relations into interpretable component–chainage patterns and, most clearly in the SJLS case, retain residual-consistent co-variation after simple trend-based null explanations are considered. The evidence supports a diagnostic spatial representation claim rather than contact-force recovery, calibrated jamming risk, or broad predictive dominance. The proposed framework provides a spatial representation basis for future rock–machine interaction diagnosis and digital-twin-oriented TBM excavation analysis.

A Exploratory descriptor–response matrix

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Table 11: Full unadjusted component–response Spearman matrix for $I_c(t)$–persistence-residual association. Correlations are exploratory descriptor diagnostics and are not adjusted for multiple testing.

Case	Component	Adv. rate	Torque	Thrust	Penetr.	Shield press.
BSLL h=1	Cutterhead	0.238	-0.286	0.048	-0.095	0.095
BSLL h=1	Front shield	0.381	-0.429	0.190	0.095	-0.095
BSLL h=1	Middle shield	-0.048	0.262	0.452	-0.167	0.143
BSLL h=1	Tail shield	-0.238	0.286	-0.048	0.095	-0.095
BSLL h=3	Cutterhead	-0.214	-0.714	-0.036	-0.214	-0.536
BSLL h=3	Front shield	0.286	-0.179	0.143	0.250	-0.750
BSLL h=3	Middle shield	0.214	0.714	0.036	0.214	0.536
BSLL h=3	Tail shield	0.179	0.750	0.107	0.143	0.571
SJLS h=3	Cutterhead	0.860	0.446	-0.669	0.669	-0.912
SJLS h=3	Front shield	0.640	0.152	-0.350	0.475	-0.745
SJLS h=3	Middle shield	-0.225	-0.449	0.478	-0.282	-0.042
SJLS h=3	Tail shield	-0.846	-0.324	0.733	-0.811	0.784

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Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The materials that can be shared are the experiment configurations, analysis scripts, figure scripts, and derived result summaries used to reproduce the reported tables and figures. These materials are available from the corresponding author upon reasonable request, subject to project data-use permissions; the raw site data are not redistributed.

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